

SANTA MARIA BUOY 46011, CA, USA

WMO: 992960

Lat: 34.956N

Lon: 121.019W

Elev: 0

StdP: 14.70

Time Zone: -8.00 (NAP)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	47.7	48.8	N/A	N/A	N/A	N/A	N/A	N/A	34.6	52.3	32.5	51.9	11.5	330	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	3.5	66.2	N/A	65.0	N/A	63.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.3	26.1	24.2	45.2	68.8	2.3	3.5	43.5	71.4	42.1	73.4	40.8	75.4	39.1	77.9		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	54.5	53.8	53.6	52.8	53.4	55.0	57.2	58.3	58.8	59.0	57.6	55.1
	DBStd	3.34	3.06	2.67	2.46	2.04	1.74	1.84	2.63	2.49	2.78	3.01	2.68	2.66
	HDD50	7	3	2	1	1	0	0	0	0	0	0	0	0
	HDD65	3375	325	313	354	366	359	301	243	209	186	188	223	308
	CDD50	2111	143	109	112	86	106	149	222	256	265	279	227	157
	CDD65	2	0	0	0	0	0	0	0	0	1	1	0	0
	CDH74	2	0	0	0	2	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.7	11.9	13.1	13.6	14.7	14.8	13.8	11.2	11.6	11.3	11.5	12.0	12.4
Precipitation	PrecAvg	14.0	2.8	3.0	2.1	1.1	0.6	0.2	0.1	0.1	0.2	0.7	1.2	2.2
	PrecMax	28.4	12.2	12.9	9.7	3.8	2.4	0.7	0.5	0.4	0.7	3.3	3.9	7.2
	PrecMin	4.6	0.2	0.2	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.1	0.2	0.2
	PrecStd	5.6	2.6	2.6	2.1	0.9	0.5	0.2	0.1	0.1	0.1	0.7	0.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	62.0	62.2	60.8	61.1	61.7	66.3	65.4	68.1	67.7	65.1	63.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	60.9	60.3	60.0	57.4	58.1	59.6	64.0	64.4	66.1	66.1	63.3	62.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	59.7	58.9	58.2	56.2	56.9	58.6	62.4	63.5	64.4	65.1	62.1	60.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	58.7	57.4	57.1	55.4	56.0	57.6	61.1	62.4	62.9	63.8	61.1	59.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	4.2	3.7	3.4	3.1	2.7	2.9	2.9	3.0	3.2	3.5	3.9	4.0
		MCDBR	5.4	4.8	4.7	4.5	3.7	3.9	3.2	3.1	3.5	4.2	5.1	4.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.313	0.324	0.339	0.347	0.354	0.352	0.357	0.344	0.347	0.346	0.321	0.319	
	taud	2.587	2.566	2.520	2.491	2.493	2.491	2.533	2.546	2.533	2.538	2.573	2.580	
	Ebn at Noon	285	294	297	298	295	294	292	295	290	281	279	276	
	Edh at Noon	24	28	31	34	34	34	33	32	31	28	24	23	
All-Sky Solar Radiation	RadAvg	930	1200	1629	2019	2221	2274	2049	1895	1717	1348	1038	823	
	RadStd	72	122	112	140	164	126	111	126	113	105	62	80	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	+0.95	N/A	-2.30	+1.31	N/A	N/A	N/A	-297	+367	+79

Nomenclature: See separate page